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TRAVEL- MUSEUM

ABSTRACT

The Travel Museum Digital Cloud center is a cloud-based solution designed to store, manage databases and showcase the rich heritage of travel and exploration. This project mainly focuses on the scalability, flexibility and accessibility of cloud computing to create a centralized digital repository for travel artifacts, documents, and multimedia content. By hosting the system on the cloud, it enables seamless access for researchers, tourists, and museum staff across the globe. Overall, this system aims to modernize cultural preservation through innovative technology, making the museums' collection more accessible and engaging in the digital age.

The project aims to:

- Digitise the collection of museum artifacts, exhibits and visitors' information.
- Enable remote access to exhibits, visitor data, and archival materials.
- Provide data analytics for curatorial improvement, visitors patterns, and recommendation system

This cloud-based transformation ensures sustainable digital preservation while enriching interactive experiences for a global audience.

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TRAVEL MUSEUM

INTRODUCTION

In modern times museums are transforming their visitors' experience by changing into digital infrastructure such as by assistance of cloud computing. Meanwhile Travel Museum traditionally relied on **physical, on-premises data storage systems** and **manual processes** to manage its vast array of digital content, historical records, visitor information and administration details. Mainly **traditional systems** involve physical hardware like servers, storage devices and networking equipment, and internal IT teams. Because traditional approach to data management focused on managing physical infrastructure for every service.

The **traditional system** began to face various crucial challenges as it grew and began to **digitize** its exhibitions and records. Due to its **limited scalability** because existing storage has become insufficient. Maintaining **offline physical servers** and **storage** required more energy and human resources. Other drawbacks such as limited scalability, high costs, accessibility challenges, and risk of data loss.

The **aim** of this project is to establish a cloud-based data management system for a Travel Museum using AWS. The system is efficiently managed, stores and analyzes the data of museums, including records of the visitors and historical artifacts. The **benefits** of integrating the data into cloud and its analytics will support personalized recommendations, analysis of visitors end and efficiency of operations. This Travel Museum is based on a digital data center (cloud based) system like AWS which offers many advantages over a traditional on-premises data center. The features like **Pay-as-you-go** model, **Automatic scaling** of up and down based on visitors' traffic, **Built-in security**, **Automatic backups and recovery**, and **cloud fronts** for better performance.

PROJECT PLAN

The Gantt chart is used to exhibit the completion of the tasks.

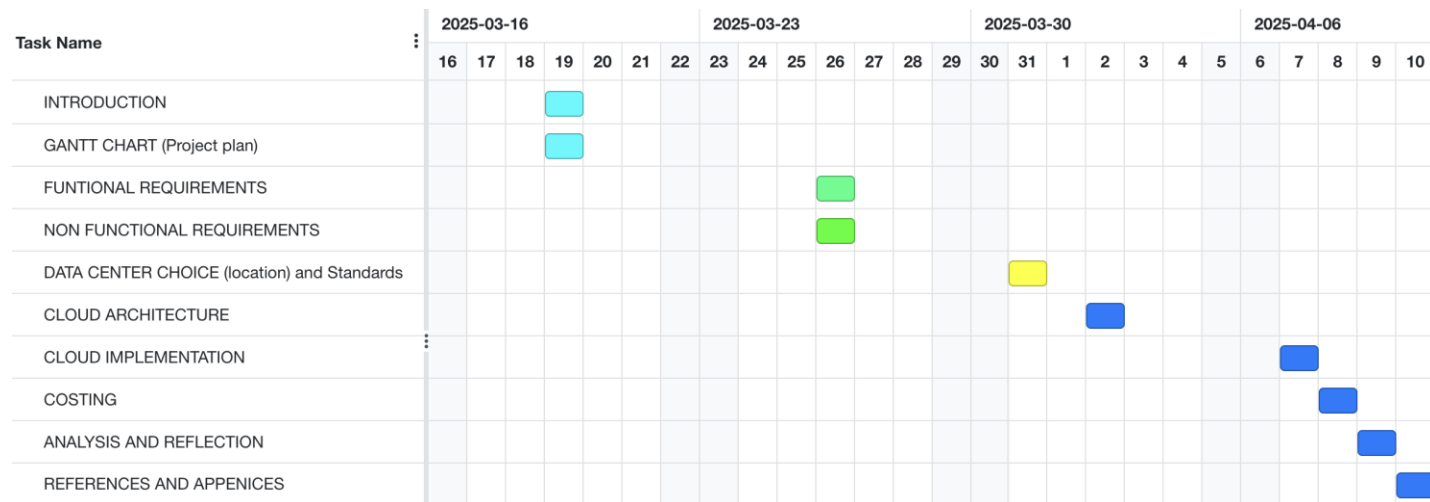


Figure 1: Gantt chart of the coursework shows milestones(<https://www.onlinegantt.com/#/gantt>)

REQUIREMENT GATHERING:

The functional requirements of my cloud-based database management application for the travel museum, based on the problems it solves (improving, data management, accessibility, and scalability).

Functional requirements of the application describe the specific functions the system should perform to solve the problems of the travel museum.

For **artifact and exhibit** data management we have the amazon **S3** (simple storage service): Store and manage multimedia files such as images, videos and documents related to artifacts and exhibits.

Visitor Management can be handled by the assistance of **amazon Cognito** it can manage user authentication and authorization, allowing for secure login and role-based access control. Supports multi-factor authentication (MFA). **DynamoDB** is to store visitor data such as personal details, preferences, and ticketing information. And to use serverless functions for processing visit interactions and ticketing actions in real-time.

User authentication and Role management is done by **amazon Cognito** again which handles user sign-ups, sign-ins, and access control. Create user pools and identity pools to manage staff members, researchers and visitors' role. Also, **IAM** (identity and access management) is for accessing control over AWS resources, ensuring only authorized users(staff/admins) can access sensitive parts of the application.

Amazon DynamoDB streams to store information and leveraging DynamoDB Streams for **real time updates**. For example, whenever a new exhibit or changes exhibit information, the system can perform change in real time. While **Amazon SNS** (Simple Notification Service) notifies users (staff or visitors) to exhibit changes in real time via email, SMS, or app notifications.

To reporting and Analytics here Amazon Quick Sight is used to create interactive dashboards and visual reports for tracking exhibit performance, ticket sales and visitor analytics.

Also, for integrating with external Systems **AWS Lambda and API Gateway:** using **AWS Lambda** in combination with API Gateway to integrate with external systems like payment gateways, CRM tools and other third-party APIs.

Amazon Event Bridge: it is used to event driven architecture to integrate external systems with museum's internal services.

The non-functional requirements of the application are used to solve the problems related to the cloud-based model.

Scalability, Amazon EC2 will automatically scale EC2 instances based on the traffic demand of the application and backend services.

AWS Lambda is used as serverless computing for dynamically scaling based on demand, where the system scales automatically without provisioning resources.

Performances, Amazon Cloud Front is a content delivery network (CDN) that can cache and deliver content (e.g. images, videos, virtual tours) with low latency to visitors globally.

Amazon Elastic Cache is for in-memory caching to speed up frequent queries and data retrieval processes, improving the application's responsiveness.

Availability and Reliability, amazon route 53 is to provide DNS service that can route traffic to healthy endpoints, ensuring high availability.

Amazon **RDS** multi-AZ deployment is to ensure database availability and fault tolerance by deploying your **RDS** instance across multiple availability zones.

Security AWS KMS (Key Management Service) is for encrypting sensitive data stored in **S3**, **RDS** or **DynamoDB**. **AWS WAF** (Web Application Firewall): Protect your application from common web exploits like SQL injection, cross-site scripting (XSS). Also, **AWS Shield** provides protection against DDoS attacks.

Usability of the application with **Amazon Cloud Front** Ensure fast delivery of content to users by caching frequently requested items and providing low-latency access for visitors globally.

AWS Amplify: A fully managed service to build, deploy and manage modern web and mobile applications optimizing the user experience across platforms.

Disaster Recovery and Backup is handled by again **amazon S3** for backups and AWS Backup, cross region replication in S3 it enables replication across regions.

Maintainability is done by Amazon **CloudWatch** for logging and monitoring to provide real time insights into the system. **For compliances** artifact provide access to compliance reports and security and privacy certification for AWS services

Cost efficiency will be managed by AWS cost explorer that will monitor and analyze AWS spending, helping us to optimize resources.

Amazon EC2 Spot Instances: Reduce costs by using EC2 spot instances for non-critical workloads(e.g.,) batch processing or analytics tasks.

Choice of data center (location) and standards (compliance)

Some of the vital key factors that affect the selection of Data center.

Geographic location: Latency and performance: I choose us-east, North Virginia (US) location close to the end for lower latency and better performance. Use CDN if users are globally connected. Physical location is vital since museums will have a global audience, a data center should geographically be closer to the primary user base to provide fast access and lower network latency. **Scalability:** Data centers in the cloud platform (like AWS, Azure, or Google Cloud) allow for scalable infrastructure. **Elasticity:** cloud resources should scale automatically based on demand, ensuring that there's always enough Compute and storage capacity when needed, especially during peak usage times like special events or online exhibitions. **Security:** data security must comply with security standards (like ISO 27001) to protect sensitive museum data, such as visitor information and digital archives. Also, data should be encrypted during transit and at rest from¹ unauthorized access. Access controls to limit who can access data and manage the infrastructure. **Disaster recovery and backup:** ensuring that the data is regularly backed up and can be restored in case of failure and multi-availability zones to replicate data loss.

Compliance and Legal consideration: to ensure the regulations, the data center needs to meet legal requirements such as GDPR. Countries have specific laws about where data can be stored and processed. Like I have used GDPR(Europe) which requires data of EU citizens to be stored within the EU. Environmental Impact most of cloud providers use energy efficient data centers dedicated to sustainability. Also, green energy is used. **Support and SLAs:** It evaluates the level of support the data center provider offers. Look for service level agreements (SLAs). Also, some providers have 24/7 support and access for constant monitoring.

Cloud Architecture: Cloud solution

Choice of type of cloud platform as solution for the problem domain and rationale for choice:

For the museum digital cloud center, a public cloud platform has been chosen as most suitable solution. I chose Amazon web services (AWS) due to its wide range of services, global availability, cost-effectiveness, and strong support for scalability, storage, security and data analytics.

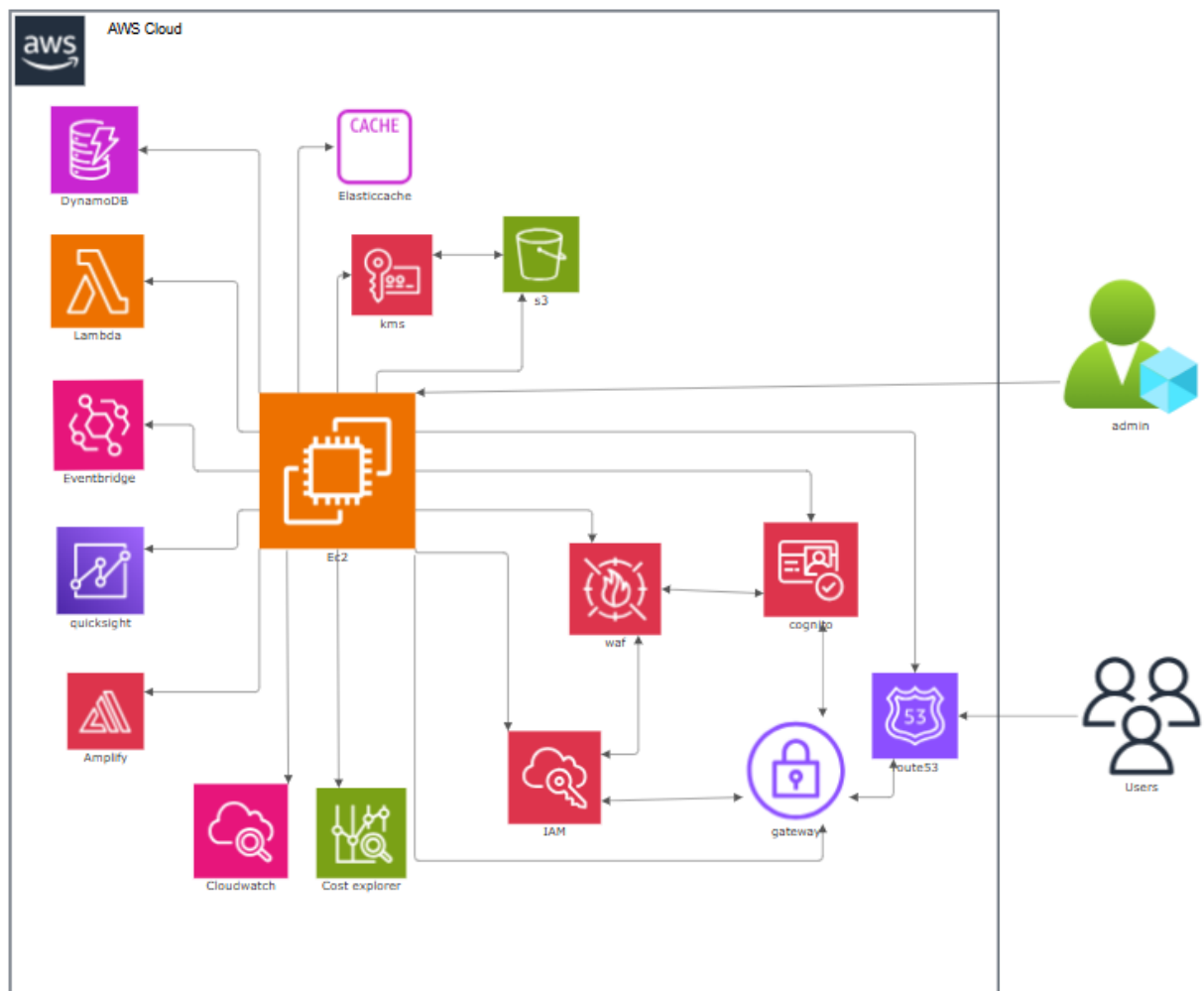


FIGURE 2 AWS ARCHITECTURE

Cloud System Architecture developed to solve the problem

The architecture of the digital cloud center is designed to provide high availability, scalability, and security while ensuring smooth access and interaction with museum's digital content. The system primarily hosted on Amazon web services (AWS) and includes the following key components:

- 1) Presentation Layer (Frontend)- Amazon S3 as a static website or through AWS Amplify. It delivers a user-friendly interface.
- 2) Application Layer (Backend Logic)- Hosted on Amazon EC2 or AWS lambda. Manage user request, handles authentication and connects with the database.
- 3) Database Layer- Amazon DynamoDB can be used for unstructured metadata or high-speed lookups.
- 4) Storage Layer- Amazon s3 used for storing the images, videos, and documents.
- 5) Content Delivery and Optimization- Amazon CloudFront (CDN) delivers content quickly to users across various geographical location with low latency.
- 6) Monitoring and Security: Aws CloudWatch is used to monitor application health, performance and usage patterns. AWS IAM ensure secure user roles and permissions. AWS WAF and SSL\TLS encryption provide an additional security layer.
- 7) Analytics and Machine Learning- Amazon QuickSight for generating visual insight from usage data

CLOUD IMPLEMENTATION

THE PROJECT REPORT.docx x Course Modules: AWS Acad... x Sandbox Environment x Workbench - Vocareum x Launch an instance | EC2 | us-east-1 console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances: us-east-1 console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances: [Option+S] United States (N. Virginia) voclabs/user3868105~u2824224@uel.ac.uk @ 3028-6742-0750

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-005bea49d8bc42caf)

[Launch log](#)

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

< 1 2 3 4 5 6 >

Create billing and free tier usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

Create Load Balancer

Create an application, network gateway or classic Elastic Load Balancer

Create Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Create AWS budget

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

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COSTING

Cost estimation using AWS calculator: -

Used AWS pricing Calculator to estimate cost of required services. Services include in the application are displayed in the screenshot below. Major cost contributions are: -

Compute (EC2, lambda)- using t3.medium and lambda has no fixed storage and set for 50 request per month.

storage (S3) 320gb per month storage for images videos and document backup. And for **RDS** 30gb per month for database and visitor.

Networking (CloudFront) will use s3 storage.

Aws cost optimization strategies like savings plans and reserved instances, autoscaling, spot instances and s3 lifecycle policies. Also, AWS support options are developer support, business support and enterprise support.

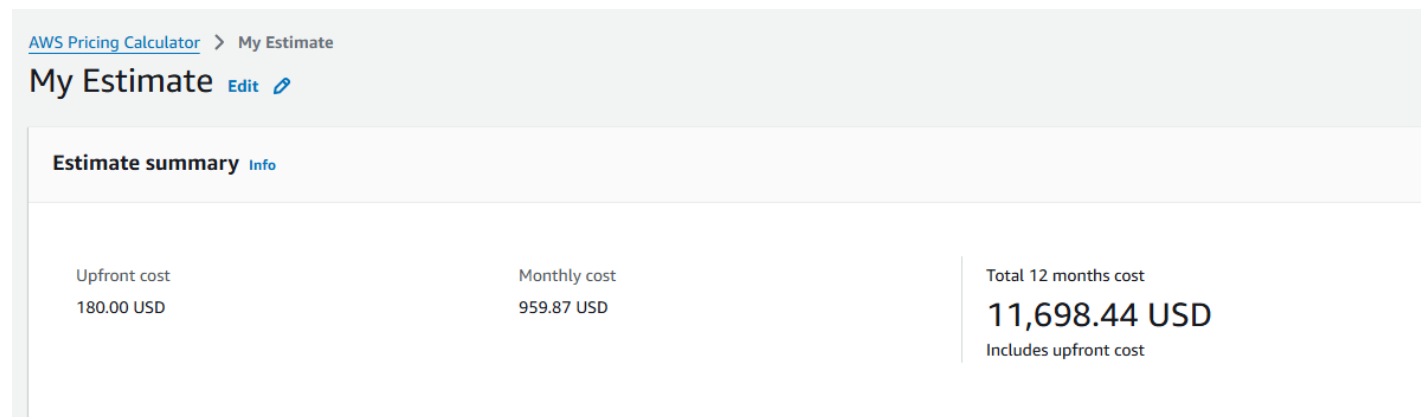


FIGURE 3: COST ESTIMATION

☐	Service Name		Status	Upfron...	Monthl...	Descrip...	Region	Config Summary
☐	AWS Lambda		-	0.00 USD	0.00 USD	-	US East (...)	Architecture (x86), Architecture (x86), Invoke Mode (Buffered), Amount of ephemeral storage allocated (512 MB), Numbe...
☐	Amazon EC2		-	0.00 USD	5.91 USD	-	US East (...)	Tenancy (Shared Instances), Operating system (Windows Server), Workload (Consistent, Number of instances: 1), Advanc...
☐	AWS Cost Explorer		-	0.00 USD	0.10 USD	-	US East (...)	Number of API requests per month (10)
☐	Amazon Cognito		-	0.00 USD	36.22 USD	-	US East (...)	Optimization Rate for Token Requests (0), Optimization Rate for App Clients (0), Advanced security features (Enabled), N...
☐	AWS IAM Access Analyzer		-	0.00 USD	80.00 USD	-	US East (...)	Number of analyzers per account (1), Number of accounts to monitor (100), Average roles per account (3), Average users ...
☐	AWS Web Application Firewall (WAF)		-	0.00 USD	90.60 USD	-	US East (...)	Number of Web Access Control Lists (Web ACLs) utilized (10 per month), Number of Rule Groups per Web ACL (1 per mo...
☐	Amazon CloudWatch		-	0.00 USD	18.85 USD	-	US East (...)	Number of Metrics (includes detailed and custom metrics) (2), GetMetricData: Number of metrics requested (10), Numbe...
☐	Amazon DynamoDB		-	180.00 USD	26.14 USD	-	US East (...)	Table class (Standard), Average item size (all attributes) (1 KB), Write reserved capacity term (1 year), Read reserved capa...
☐	Amazon Simple Notification Service (SNS)		-	0.00 USD	101.98 USD	-	US East (...)	Requests (5 million per month), HTTP/HTTPS Notifications (million per month), EMAIL/EMAIL-JSON Notifications (5 mil...
☐	Amazon Simple Storage Service (S3)		-	0.00 USD	7.13 USD	-	US East (...)	S3 Standard storage (310 GB per month), PUT, COPY, POST, LIST requests to S3 Standard (100), GET, SELECT, and all oth...

FIGURE 4: AWS SERVICES (a)

☐	Service Name		Status	Upfron...	Monthl...	Descrip...	Region	Config Summary
☐	AWS Amplify		-	0.00 USD	1.19 USD	-	US East (...)	Duration of each request (in ms) (500), Number of build minutes (30 per month), Data stored per month (5 GB), Data ser...
☐	Amazon EventBridge		-	0.00 USD	0.00 USD	-	US East (...)	Size of the payload (1 KB), Number of AWS management events (1 per month), Number of AWS opt-in data events (1 pe...
☐	Amazon ElastiCache		-	0.00 USD	91.25 USD	-	US East (...)	Instance type (cache.m5.xlarge), Cache Engine (Redis), Nodes (0), Utilization (On-Demand only) (100 %Utilized/Month), ...
☐	Amazon Route 53		-	0.00 USD	500.50 USD	-	US East (...)	Hosted Zones (1), Additional Records in Hosted Zones (1)

FIGURE 5: AWS SERVICES (b)

Analysis and Reflection

To maintain and manage the evolution of the Travel Museum Digital Cloud Center system, several points are implemented to address changing business needs, regulations, and government rules.

1. System Maintenance and updates: regular monitoring and performance optimization, utilize AWS CloudWatch and other monitoring tools to keep track of the system's performance.
2. Managing System Evolution: scalability as the number of visitors to the museum grows or business requirements change, the system must be able to scale efficiently. Auto scaling, and Amazon DynamoDB's on-demand mode for handling traffic spikes without manual intervention. Aws Lambda is serverless to allow for easier updates, reduces the risk of breaking changes, and enables faster rollouts of new features.
3. Responding to evolving business needs: agility and adaptability- to quickly adapt to new business requirements, implement an agile development process. This includes working closely with business stakeholders and museum curators to gather feedback, identify new opportunities, and prioritize changes or features.
4. Regulatory Compliance and government rules: data privacy and security: as regulations around data privacy (e.g., GDPR in EU)
5. Regular Audits: stay ahead of regulatory requirements by conducting regular security audits and ensuring that your architecture adheres to both local and international laws.
6. Collaboration with Stakeholders: Feedback Loops: establish a regular feedback loop with both users and internal stakeholders (museum staff, curators) to understand changing business goals and regulations.

Conclusion

The Travel Museum digital data center represents a modern, scalable and intelligent platform for digital transformation within the cultural heritage sector. By leveraging the powerful ecosystem of AWS services such as DynamoDB for scalable data management, Aws lambda for serverless compute and ec2 for the hosting and enhancing the visitor experience while ensuring operational efficiency and long-term sustainability.

Throughout the project, emphasis was placed on creating a system that is not only robust and secure but also adaptive to changing business needs, user expectations and regulatory frameworks like GDPR and needs of maintaining the integrity of the organization.

Moreover, the collaborative approach engaging with curators, developers, and stakeholders ensures that the system evolves in harmony with museum's goals, government policies, and technological advancements.

In essence, the TRAVEL MUSEUM DIGITAL CLOUD CENTER is more than just a digital repository –it's a dynamic platform that celebrates cultural heritage while embracing the future of cloud innovation, data-driven personalization, and secure digital engagement.

REFERENCES

- Amazon Web Services (AWS), 2025. Available at: <https://aws.amazon.com/>
- European Union, 2016. *General Data Protection Regulation (GDPR)*. [online] Available at: <https://gdpr.eu/> [Accessed 3 April 2025].
- Amazon Web Services (AWS), 2025. *Service Level Agreements (SLAs)*. [online] Available at: <https://aws.amazon.com/legal/service-level-agreements/> [Accessed 3 April 2025].
- Amazon Web Services (AWS), 2025. *AWS Pricing Calculator*. [online] Available at: <https://calculator.aws/> [Accessed 3 April 2025].

APPENDIX A: USEFUL RESOURCES

- Simple Monthly Calculator: <https://calculator.s3.amazonaws.com/index.html>
- Online AWS Architecture Diagram Tool (visual-paradigm.com)
- Gantt Chart -<https://www.onlinegantt.com/#/gantt>